

## Task 3 – Mini Project Proposal

### Enhancing SIEM Threat Detection and SOC Automation Using AI Agents

AI/ML Intern – AI Agents Domain

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#### 1. Introduction

Modern Security Operations Centers (SOCs) face continuously growing alert volumes, complex threats, and limited analyst bandwidth. Traditional SIEM platforms depend heavily on static rules, manual investigation, and human-driven incident triage. This leads to high fatigue, slow response time, and missed threats. To address these challenges, this proposal introduces an AI-Agent Driven SIEM Enhancement Framework that improves threat detection, alert correlation, and automated SOC response using intelligent agent workflows.

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#### 2. Problem Statement

SIEM tools generate thousands of alerts daily from logs, networks, endpoints, IAM systems, firewalls, and cloud services.

However, challenges include:

High false positives

Static rules unable to detect modern attacks

Manual triage takes hours

Analysts overwhelmed by repetitive tasks

Slow threat containment

AI agents can analyze patterns, correlate signals, and automate decisions, significantly reducing analyst workload.

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### 3. Proposed Solution – AI Agent-Enhanced SIEM

The solution integrates autonomous AI agents into the SIEM pipeline:

#### a) Detection Agent

Processes logs in real-time

Applies the 20 SIEM rules (failed logins, brute force, privilege escalation, malware behavior, abnormal network flows, unusual file access, suspicious IPs, data exfiltration, etc.)

Enhances rules with ML-based anomaly detection

## b) Correlation Agent

Automatically connects evidence from multiple logs

Builds incident stories (e.g., login → privilege escalation → lateral movement)

Reduces false positives

## c) Response Agent

Suggests actions or executes automated responses:

Blocking an IP

Resetting a user session

Isolating a host

Creating a ticket

Sending alerts to SOC dashboard

## d) Analyst Assistant Agent

Explains alerts

Summarizes logs

Recommends mitigation steps

Helps junior analysts understand incidents

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#### 4. Application of SIEM's 20 Core Rules

This system utilizes and enhances SIEM rules such as:

1. Excessive failed logins
2. Brute-force attacks
3. Privilege escalation
4. Suspicious admin activity
5. New user creation anomalies
6. Disabled security tools
7. Malware hash detection
8. Unusual outbound traffic

9. Data exfiltration behavior
10. Port scanning detection
11. Large file transfers
12. Login from unusual locations
13. VPN abuse
14. Lateral movement
15. Unauthorized file access
16. Suspicious PowerShell activity
17. Firewall rule changes
18. Registry changes
19. Failed authentication spikes
20. High-severity alerts correlation

AI agents continuously learn from past incidents, improving accuracy beyond static rule matching.

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## 5. High-Level Architecture

Log Sources → SIEM → AI Detection Agent → Correlation Agent → Response Agent → SOC Dashboard

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Analyst Assistant Agent (LLM-powered)

AI agents sit on top of the SIEM and enhance it without replacing existing infrastructure.

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## 6. Benefits of the Proposed System

50–70% reduction in false positives

Faster incident response (seconds instead of hours)

Automated triage of routine alerts

Lower analyst fatigue

Better detection of zero-days & anomalies

Continuous learning system

Explainable alerts, helping new analysts understand patterns

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## 7. Conclusion

Integrating AI agents into SIEM transforms reactive security operations into proactive, automated, and intelligent defense systems. The proposed framework improves threat detection accuracy, accelerates SOC workflows, and supports analysts with real-time insights. By applying the 20 core SIEM rules and enhancing them with AI-driven analysis, the system offers a scalable and modern solution for next-generation cybersecurity operations.